THE TRIANGLE DECOMPOSITION PROPERTY IS TWO OUT OF THREE AND STABLE UNDER DIRECT SUMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11991-12028, following the statement at 11977-11989.

Editorial note. No omissions: both displayed diagrams are shown; the remaining cases of (1) are by rotation of triangles (translation mirror, established note practice).

Allowed standard prerequisites (editorial).

- Orthogonality forces uniqueness. lemma-pre-prepare-adjoint (same chapter, cited at ~11996). Hom(A_1, A_2) = Hom(A_1, X_2) gives the unique commuting morphism A_1 -> A_2.
- Nine-square extension machinery. proposition-9 (same chapter, cited at ~12005). Provides the four-row extension diagram of triangles.
- Orthogonal subcategories and split triangles. lemma-orthogonal-triangulated (cited at ~12014) and lemma-split (cited at ~12024), both same chapter. Closes A-perp under cones in triangles and handles part (2).

Source reference. Original lemma 0FXC: sources/stacks-derived/pinned-original-source.tex, source lines 11955–11955. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CQQ: sources/stacks-derived/pinned-original-source.tex, source lines 11900–11900. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QT: sources/stacks-derived/pinned-original-source.tex, source lines 917–917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 05R0: sources/stacks-derived/pinned-original-source.tex, source lines 1248–1248. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let $\mathcal{A}$ be a full triangulated subcategory of $\mathcal{D}$. For an object X of $\mathcal{D}$ consider the property $P(X)$: there exists a distinguished triangle $A \to X \to B \to A[1]$ in $\mathcal{D}$ with A in $\mathcal{A}$ and B in $\mathcal{A}^\perp$.*

(1) *If $X_1 \to X_2 \to X_3 \to X_1[1]$ is a distinguished triangle and P holds for two out of three, then it holds for the third.*
(2) *If P holds for X_1 and X_2, then it holds for $X_1 \oplus X_2$.*

Proof. Let $X_1 \to X_2 \to X_3 \to X_1[1]$ be a distinguished triangle and assume P holds for X_1 and X_2. Choose distinguished triangles

$$A_1 \to X_1 \to B_1 \to A_1[1] \quad \text{and} \quad A_2 \to X_2 \to B_2 \to A_2[1]$$

as in condition P. Since $\mathrm{Hom}(A_1, A_2) = \mathrm{Hom}(A_1, X_2)$ by Lemma 0CQQ there is a unique morphism $A_1 \to A_2$ such that the diagram

$$\begin{array}{ccc} A_1 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ A_2 & \longrightarrow & X_2 \end{array}$$

commutes. Choose an extension of this to a diagram

$$\begin{array}{ccccc} A_1 & \longrightarrow & X_1 & \longrightarrow & Q_1 & \longrightarrow & A_1[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A_2 & \longrightarrow & X_2 & \longrightarrow & Q_2 & \longrightarrow & A_2[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A_3 & \longrightarrow & X_3 & \longrightarrow & Q_3 & \longrightarrow & A_3[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A_1[1] & \longrightarrow & X_1[1] & \longrightarrow & Q_1[1] & \longrightarrow & A_1[2] \end{array}$$

as in Proposition 05R0. By TR3 we see that $Q_1 \cong B_1$ and $Q_2 \cong B_2$ and hence $Q_1, Q_2 \in \mathrm{Ob}(\mathcal{A}^\perp)$. As $Q_1 \to Q_2 \to Q_3 \to Q_1[1]$ is a distinguished triangle we see that $Q_3 \in \mathrm{Ob}(\mathcal{A}^\perp)$ by Lemma 0FXC. Since $\mathcal{A}$ is a full triangulated subcategory, we see that A_3 is isomorphic to an object of $\mathcal{A}$. Thus X_3 satisfies P. The other cases of (1) follow from this case by translation. Part (2) is a special case of (1) via Lemma 05QT. □